SMARTBRIDGE / SKILL WALL

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent Python/HTML/CSS indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Backend modules, templates, static files and tests are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as prompt, topic, summary and learning_path are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as generate_text and endpoint handlers describe their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are used where useful; more detailed comments can be added later.
6	Modular Design	Code is split into reusable functions/modules	Yes	Q&A;, explanation, quiz, summary and learning-path logic are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common Gemini generation is centralized in gemini_client.py.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and API/runtime errors are handled with clear responses.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	gemini_client.py	Python, Google GenAI SDK	Q&A;, quiz, summary and learning-path modules	High
2	qna.py	Python / FastAPI	/qa and backward-compatible API route	High
3	explanation_module.py	Python, Transformers, LaMini-Flan-T5	/explain endpoint	Medium
4				

Reusability Level (High /

Pytg BT UIF1 7. flowhon, Transformers,